Dipayan Chatterjee

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SUMMARY

Detail-oriented Data Scientist with expertise in statistical analysis, data visualization, machine learning, deep learning, and Generative AI to support business decision-making. Experienced in building predictive models, including a customer churn prediction system that delivers key insights into churn patterns, as well as RAG-powered document analysis and Q&A solutions. Brings a strong problem-solving mindset, a solid statistical foundation, the ability to uncover insights and building end-to-end data science solutions aligned with business goals.

SKILLS

Technical Skills: Python — Machine Learning — Deep Learning — NLP — NLTK — Tensorflow — Transformers — LangChain — LangGraph — Generative AI — RAG — AWS Cloud — Predictive Analytics — Pandas — NumPy — Matplotlib — Seaborn — Scikit-Learn — SQL — Power BI — MS Excel — Statistical Modeling — Docker

Analytical & Domain Skills: Statistical Analysis — Hypothesis Testing — Data Cleaning — Data Interpretation — Business Intelligence — Problem-Solving — Data-Driven Decision Making — Research & Forecasting

EXPERIENCE

iStreet Network Limited — Mumbai, India
ML Engineer — March, 2026 – Present

- Extracted and analyzed banking application health monitoring data from OpenSearch DB by building end-to-end preprocessing, EDA, feature engineering pipelines, and implemented an Isolation Forest–based anomaly detection system to identify context aware abnormal behaviours.

Magic9 Media & Analytics Pvt. Ltd — Mumbai, India
Data Science Analyst — July, 2025 – March, 2026

- Designed and implemented an audience segmentation and program affinity modeling framework on TV viewership and panel data, enabling data-driven media planning and targeted content strategies in association with BARC India.
- Enhanced BARC India's panel churn prediction by performing efficient data preparation, feature engineering, and EDA, uncovering high-impact behavioural and demographic insights.

Hands-on Projects

Customer Churn Prediction in the Banking Sector (Web Deployment) — (March, 2025 - April, 2025)

Summary: Covers data cleaning, visualization, feature analysis, and predictive modeling using an Artificial Neural Network (ANN) to classify customer churn through a web application.

- Engineered an ANN model achieving 87% training accuracy and 86% validation accuracy with balanced loss values,indicating strong generalization.
- Identified key churn drivers including higher churn proportion among female customers and age-related patterns, with median age 45 showing higher churn tendency.

RAG-Powered Document Analysis and Q&A — (September, 2025)

Summary: Built a complete Retrieval Augmented Generation (RAG) pipeline for interactive conversational Q&A over PDF documents.

- Processed chunked raw text using LangChain, then created semantic vector embeddings with Sentence-Transformers.
- Indexed the embeddings in ChromaDB to enable efficient retrieval of highly relevant and contextually grounded content.
- Deployed Llama3 with prompt engineering and map-reduce summarization chains for accurate, grounded responses.

Other notable projects: Self RAG and Corrective RAG applications, Bank Customer Salary Estimation (Regression), Hamlet Next-Word Prediction (NLP), Kindle Review Sentiment Analysis (NLP & Classification)

EDUCATION

M.Sc. - Statistics
Pondicherry University (Pondicherry)
CGPA: 9.32
2023 – 2025

B.Sc. - Statistics
Calcutta University (Kolkata)
CGPA: 8.32
2020 – 2023

WBCHSE (XII)
Patha-Bhavan (Kolkata)
93.4%
2018 – 2020

CERTIFICATIONS

- Data Science using Python
- Predictive Analytics

PUBLICATIONS

Bird Species Classification Using MobileNetV2 and CNN: A Comparative Study with Local Web Deployment — IOSR Journal of Computer Engineering (IOSR-JCE)